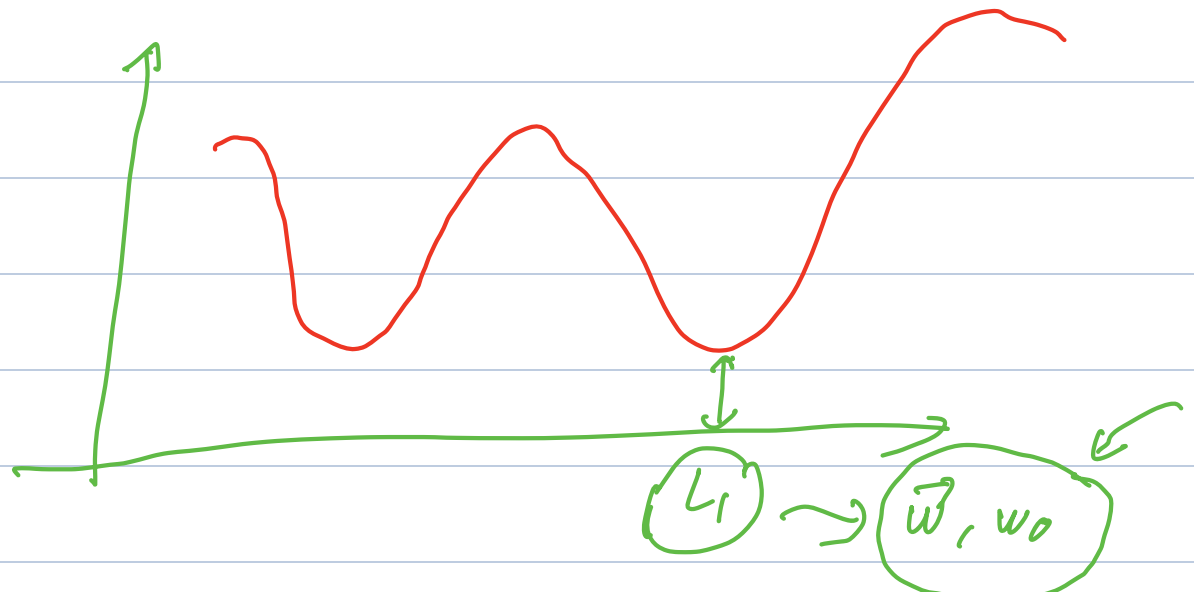
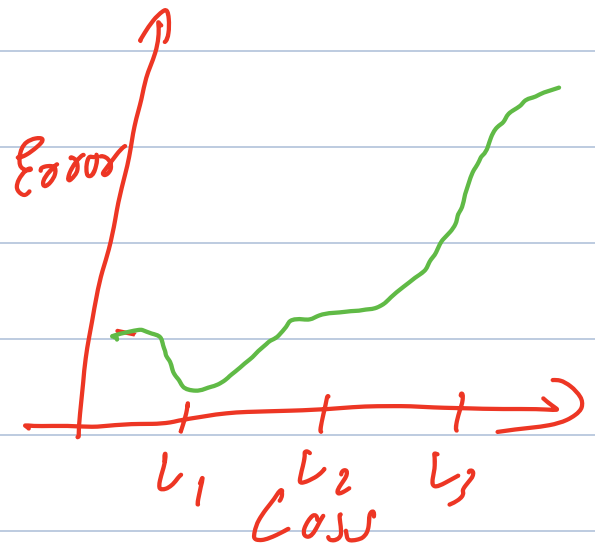
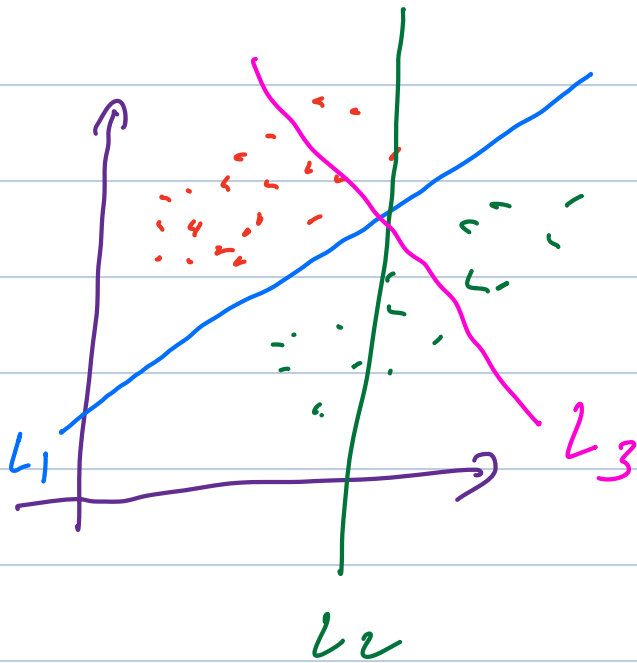


Need for Calculus in ML

- Basic Intuition of Classification
- How to solve optimization prob
- Define classification prob mathematically
- Rel b/w dist & gain func
- Functions / some more func
- Limits



$$w^T x + w_0 = 0$$

$$x_1, x_2 \rightarrow w_1 x_1 + w_2 x_2 + w_0 = 0$$

$$w_1, w_2, w_0$$

$$w_1 \rightarrow [-10 \text{ to } 10] \rightarrow 0.1$$

$$[-10, -9.9, -9.8, \dots] [0] [0.1, 0.2, 0.3, \dots, 10]$$

$$w_1 \rightarrow 201$$

$$w_2 \rightarrow 201$$

$$w_0 \rightarrow 201$$

$$\frac{201}{w_1} * \frac{201}{w_2} * \frac{201}{w_0}$$

$$-10, -10, -10$$

$$-10, -10, -9.9$$

$$-10, -10, -9.8$$

$$1 \text{ operation} \rightarrow 10^{-6} \text{ sec}$$

$$(201)^3 \text{ oper.} \rightarrow (201)^3 * 10^{-6} \text{ sec}$$

How to solve for this optimization?

→ Use Gradient Descent

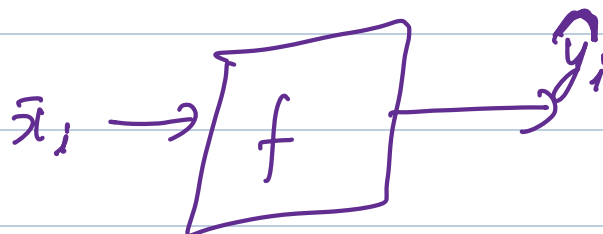
- Maxima Minima
- calculus in multivariable
- single
- derivative, slope, tangent
- Limit, continuity, differentiability

Calculus in Mathematical terms

$$D : \{ \underbrace{(x_i, y_i)}_{\text{features}} \overset{\text{label}}{\rightarrow} \quad \begin{aligned} x_i &\in \mathbb{R}^d \\ y_i &\in \{-1, +1\} \end{aligned}$$

Age x_1 #Bedrooms x_2 ~ Price y_i

Goal: Find a function $f(x)$



$$\hat{y}_i \approx y_i$$

$$f(x_i) = \hat{y}_i$$

↓
predicted value

# Exp	Salary
1	10K
2	20K
3	31K
4	40K
5	→ ? → 51K

$$\begin{aligned}
 & f(x) \\
 & \downarrow \\
 & 10K \times \# \text{ Exp} \\
 & 10000 \times 5 \\
 & = 50K
 \end{aligned}$$

$$G(D, \vec{w}, w_0) = \left(\frac{\vec{w}^T x_i + w_0}{\|\vec{w}\|} \right) \cdot y_i$$

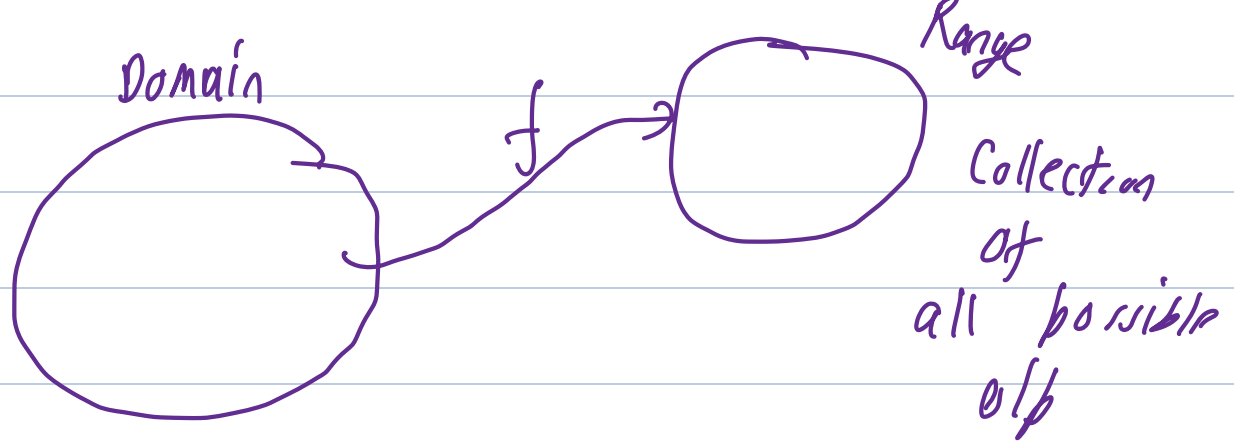
↓

we want to optimize

$$\textcircled{1} \quad G(D, \vec{w}, w_0) = \sum_{i=1}^n g(D, \vec{w}, w_0)$$

function

$$\begin{array}{ccc}
 f(x) & = & y \\
 \uparrow & & \uparrow \\
 \text{i/p} & & \text{o/p}
 \end{array}$$



$$y = f(x) = x^2$$

$$2 \rightarrow 4$$

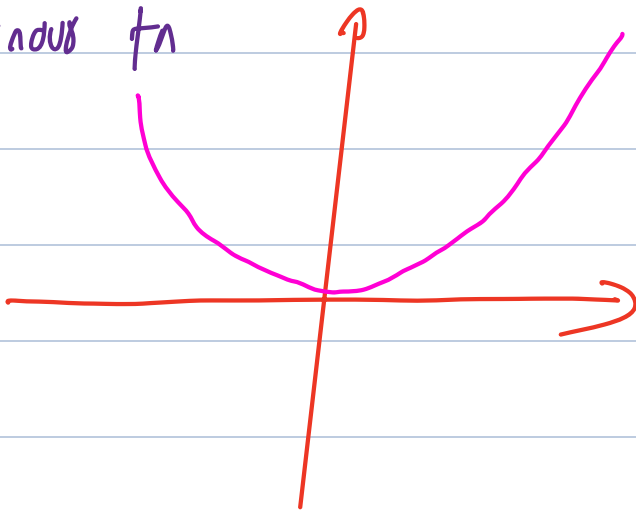
$$-2 \rightarrow 4$$

$$-\infty \text{ to } +\infty \rightarrow \text{i/p}$$

$$0 \text{ to } \infty \rightarrow \text{o/p}$$

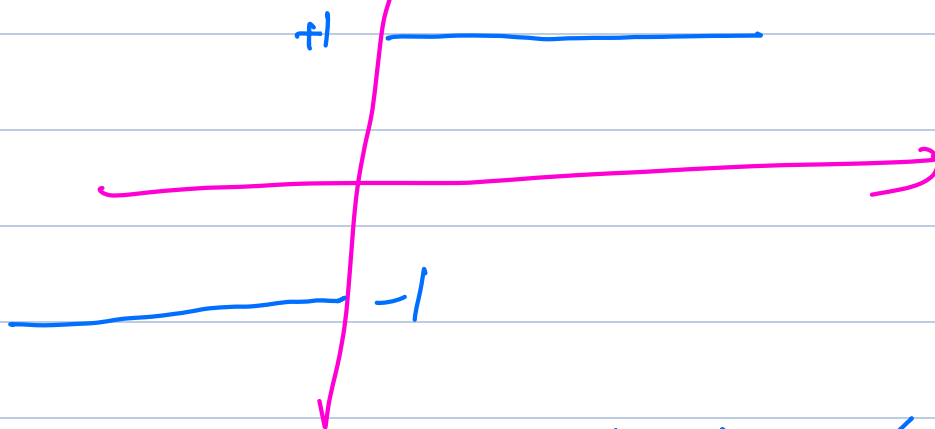
Limit

Continuous fn

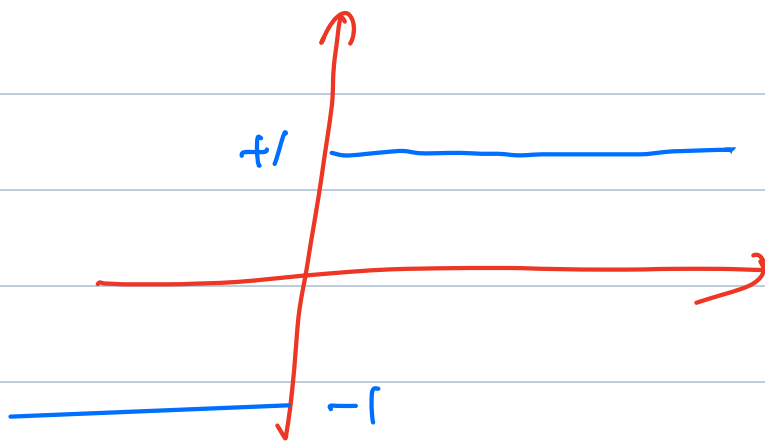


$$f(x) = x^2$$

Dis continuous function



$$y = f(x) = \begin{cases} -1 & x < 0 \\ +1 & x > 0 \\ 0 & x = 0 \end{cases}$$



at $x=0$

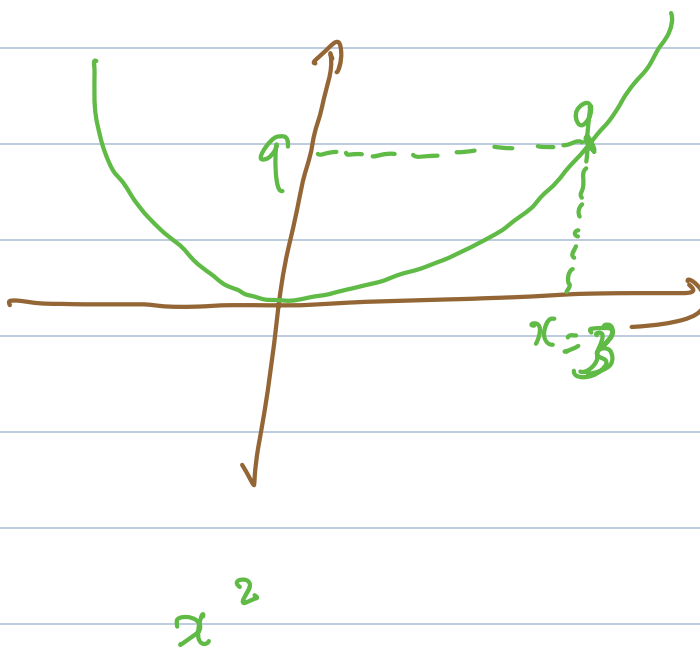
$$0^+ \rightarrow 0.1, 0.01, 0.001, 0.0001$$

Right hand Limit

$$\lim_{x \rightarrow 0^+} f(x) = 1$$

Left hand limit

$$\lim_{x \rightarrow 0^-} f(x) = -1$$



$$\lim_{x \rightarrow 3^+}$$

$$\lim_{x \rightarrow 3^-}$$

$$(3.0000001)^2$$

$$= 9.0000006000000001$$

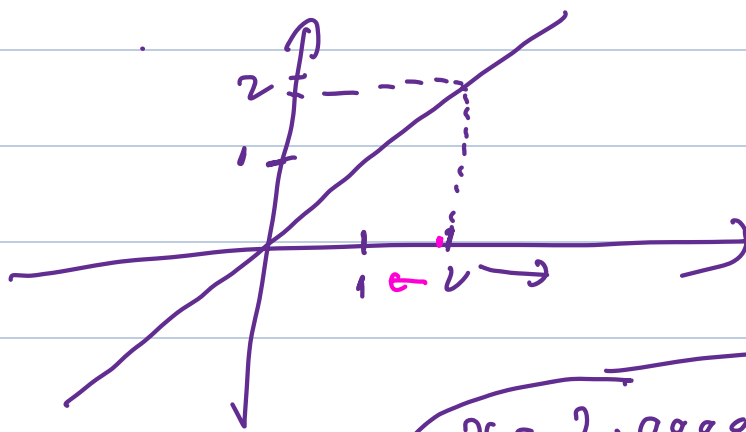
$$(2.9999999999999999)^2$$

$$= 8.9999999999999996$$

Some Imp function

Func





$$y = x$$

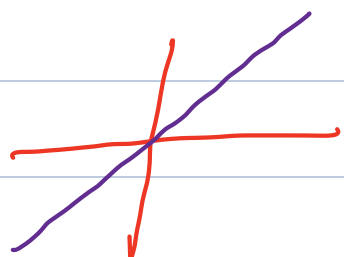
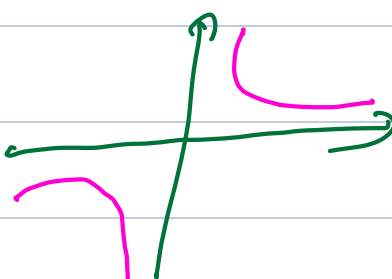
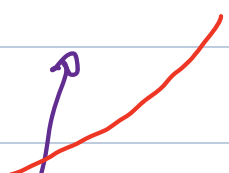
$$x = 2.000000001$$

$$y = 2.000000001$$

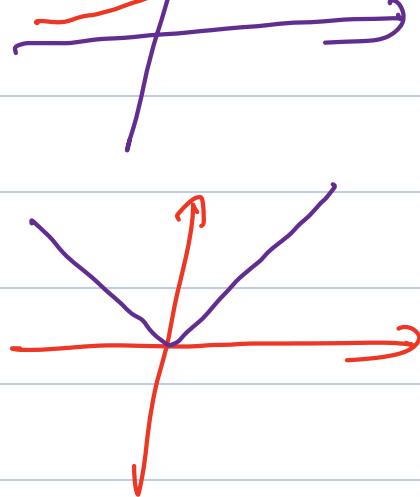
$$x = 1.999999999$$

$$y = 1.999999999$$

Some Imp function

Func	Domain	Range	Continuous	Plot
① $y = x$	$(-\infty, +\infty)$	$(-\infty, +\infty)$	Continuous	
② $y = \frac{1}{x}$	$(-\infty, \infty)$	$(-\infty, \infty)$	Discontinuous	
③ $y = e^x$	$(-\infty, \infty)$	$(0, \infty)$	Continuous	

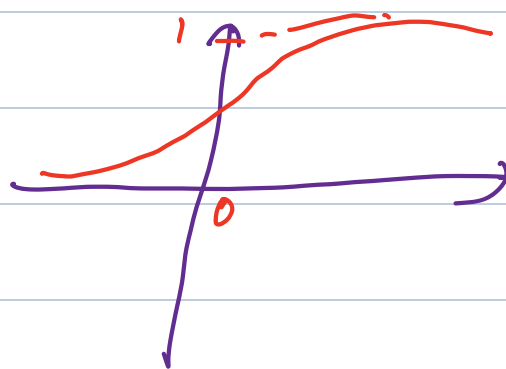
④ $y = |x|$ $\mathbb{R}$ $\mathbb{R}^+$ $(-\infty, \infty)$ $[0, \infty)$ (continuous)



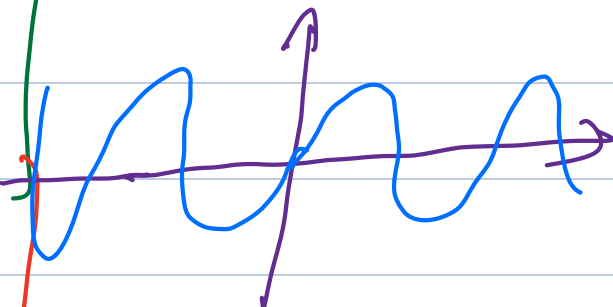
⑤ $y = \ln(x)$ $(0, \infty)$ $(-\infty, \infty)$ (continuous)



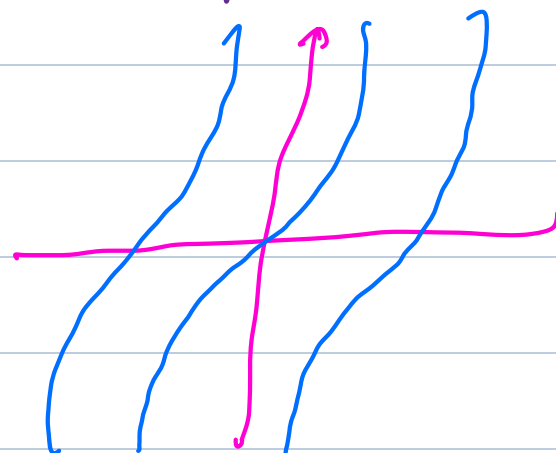
⑥ $y = \frac{1}{1+e^{-x}}$ $(-\infty, \infty)$ $(0, 1)$ (continuous)



⑦ $y = \sin(x)$ $(-\infty, \infty)$ $[-1, 1]$ (continuous)

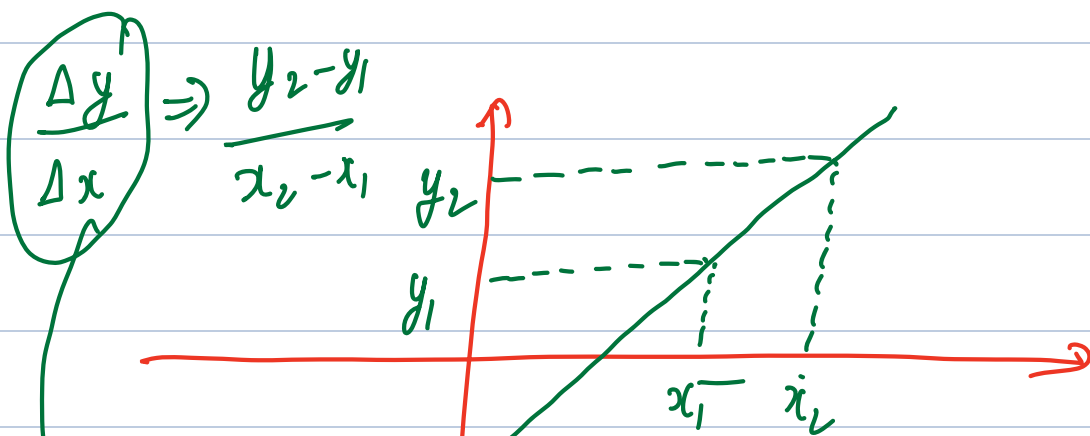
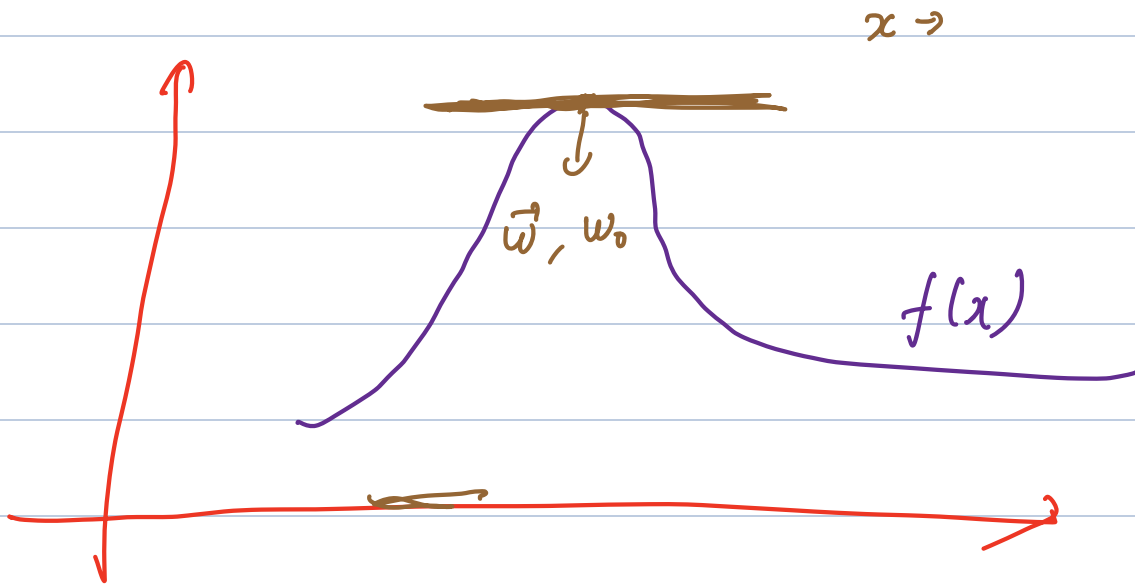


⑧ $y = \tan(x)$ — (Discontinuous)



$$\operatorname{argmax}_{\vec{w}, w_0} G(D, \vec{w}, w_0)$$

$$\operatorname{argmax}_{\vec{w}, w_0} \sum_{i=1}^n y_i \cdot \frac{w^T x_i + w_0}{\|w\|} \rightarrow \text{optimization problem}$$



$$x_1 = 5.0000000$$

$$x_2 = 5.00000001$$

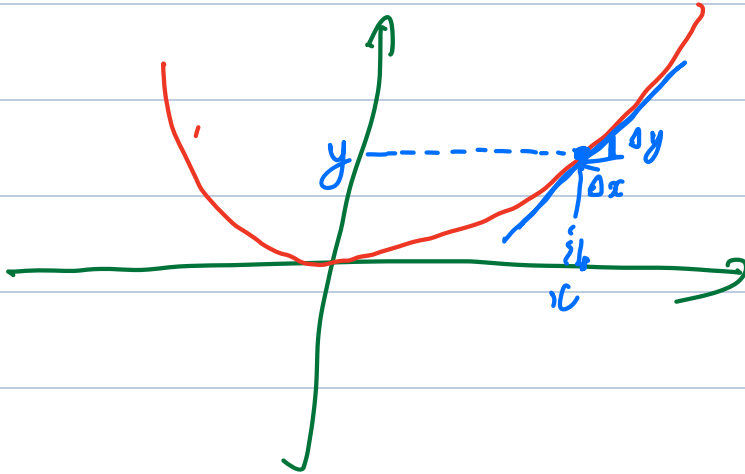
$$\frac{dy}{dx}$$

$$\Delta x \rightarrow 0$$

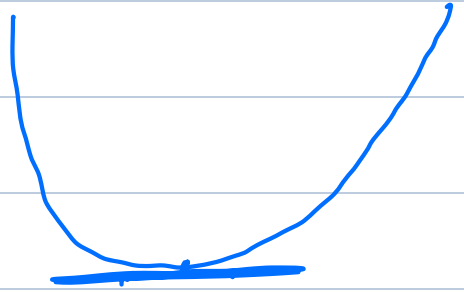
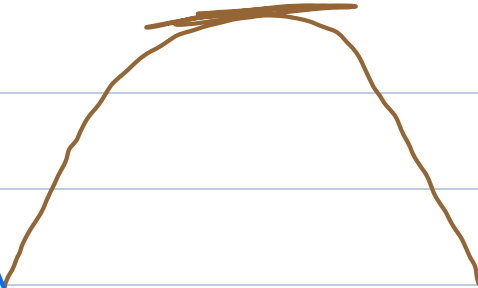
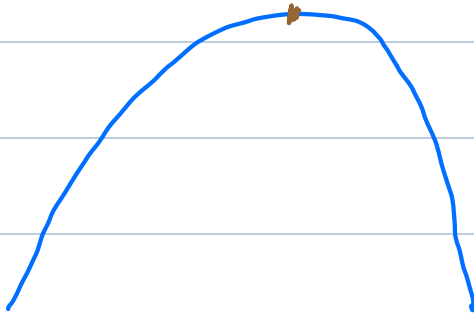
$$\rightarrow 0.0000000$$

$$\frac{d f(x)}{dx}$$

$$y = x^2$$



ML



minima

$$x'(4) = 0$$

$$x'(7) = 0$$

$$x'(5) = ?$$



$$\lim_{s \uparrow} = 1$$

$$5.00000001$$

$$\lim_{s \downarrow} = 0$$

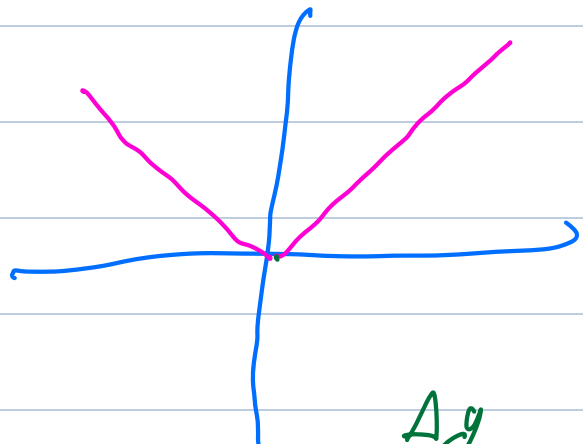
$$4.99999999$$

$$\frac{f(5.00000001) - f(4.99999999)}{0.000000002}$$

$$\frac{1 - 0}{0.000000002} \approx \infty$$

$$02 \Rightarrow f(x) = |x|$$

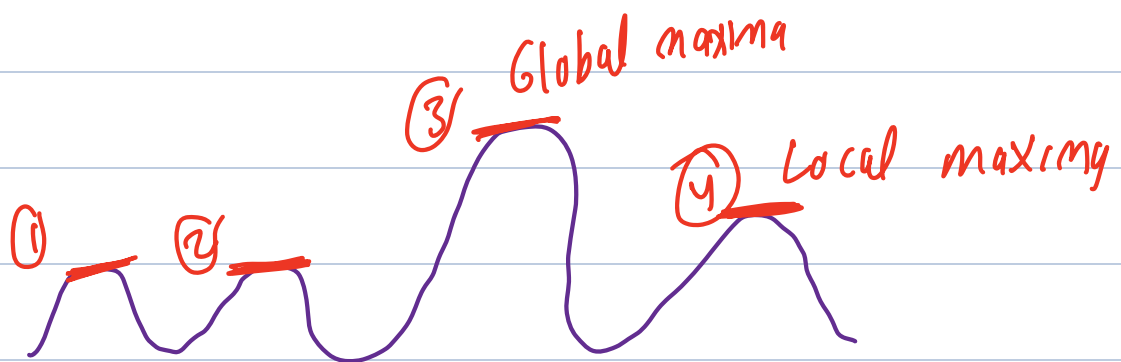
$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$



$$\frac{\Delta y}{\Delta x} = \frac{1}{1}$$

$$\lim_{x \rightarrow 0^+} f'(x) = 1 \quad x > 0$$

$$\lim_{x \rightarrow 0^-} f'(x) = -1 \quad x < 0$$



$$\Rightarrow f(x) = x^2 \quad \Rightarrow 2x$$

$\swarrow$ $\searrow$
 $(x+\Delta x)^2$
 $\boxed{x+\Delta x}$ $\frac{dy}{dx} \Rightarrow \frac{\Delta y}{\Delta x}$

$$\Rightarrow \frac{(x+\Delta x)^2 - x^2}{(x+\Delta x) - x}$$

$$\Rightarrow \frac{\cancel{x^2} + \Delta x^2 + 2x\Delta x - \cancel{x^2}}{\Delta x}$$

$$\Rightarrow \frac{\Delta x^2 + 2x \Delta x}{\Delta x}$$

$$\Rightarrow \frac{\cancel{\Delta x} (\Delta x + 2x)}{\cancel{\Delta x}}$$

$$\Rightarrow 2x + \Delta x$$



$$\Delta x \sim 0$$

$$\frac{dx^3}{dx} \Rightarrow 3x^2$$

$$\frac{dx^n}{dx} \Rightarrow n x^{n-1}$$

$$\frac{dc}{dx} = 0$$

$$\frac{d \sin x}{dx} = \cos x$$

$$\frac{d \cos x}{dx} = -\sin x$$

$$\frac{d \log(x)}{dx} = \frac{1}{x}$$

Sum Rule

$$\frac{d}{dx} [f(x) + g(x)] = f'(x) + g'(x)$$

eg: $x^2 + 7x + 12$

$\Downarrow$

$$2x + 7$$

Product Rule

$$\frac{d}{dx} [f(x) \cdot g(x)]$$

$$\Rightarrow f(x) \cdot g'(x) + g(x) \cdot f'(x)$$

$$\rightarrow (x+3)(x+4)$$

$$x^2 + 4x + 3x + 12$$

$$x^2 + 7x + 12$$

$$\rightarrow \boxed{2x + 7}$$

$$(x+3) \frac{d}{dx} (x+4) + (x+4) \frac{d}{dx} (x+3)$$

$$(x+3) \cdot 1 + (x+4) \cdot 1$$

$$x+3 + x+4$$

$$\boxed{2x+7}$$

$$\Rightarrow x \cdot \log(x)$$

$$x \cdot \frac{d \log(x)}{dx} + \log(x) \cdot \frac{dx}{dx}$$

$$\cancel{x} \cdot \frac{1}{\cancel{x}} + \log(x)$$

$$1 + \log(x)$$

$\Rightarrow$ Quotient Rule

$$\frac{d}{dx} f(x)/g(x) \Rightarrow f(x) \cdot g'(x)$$

↓

$$\Rightarrow \frac{g(x) \cdot f'(x) - g'(x) \cdot f(x)}{g(x)^2}$$

Ex: $d \left[\frac{\log(x)}{x} \right]$

$$\frac{x \cdot \frac{d \log(x)}{dx} - \frac{dx}{dx} \cdot \log(x)}{x^2}$$

$$\frac{1 - \log x}{x^2}$$

$\Rightarrow$ Chain Rule

$$\frac{d}{dx} f(g(x)) = f'(g(x)) \cdot g'(x)$$

ex: $e^{5x^2+2} \Rightarrow e^{5x^2+2} \cdot \frac{d(5x^2+2)}{dx}$

$$\Rightarrow e^{5x^2+2} (10x)$$

$\Rightarrow$ $f(x) = \frac{1}{1+e^{-x}}$

$\downarrow$

$$f(x) (1-f(x))$$

$$\frac{(1+e^{-x}) \cdot \frac{d1}{dx} - 1 \frac{d(1+e^{-x})}{dx}}{(1+e^{-x})^2}$$

$$\Rightarrow \frac{-e^{-x} \cdot \frac{d(-x)}{dx}}{(1+e^{-x})^2}$$

$$\Rightarrow \frac{e^{-x}}{(1+e^{-x})^2}$$

$$\Rightarrow \frac{1}{(1+e^{-x})} \cdot \frac{e^{-x}}{(1+e^{-x})}$$

$$\Rightarrow \frac{1}{1+e^{-x}} \cdot \left(1 - \frac{1}{1+e^{-x}} \right)$$

$$f(x) (1 - f(x))$$

$\Rightarrow$ Maxima & minima

$$f(x) = x^2$$

$$f'(x) = 2x$$

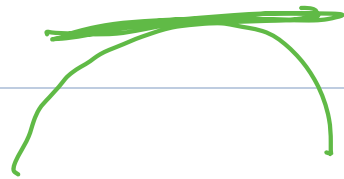
$$f''(x) = 2 \rightarrow \text{+ve}$$



$$f(x) \rightarrow f'(x) + f''(x) > 0 \rightarrow \text{mining}$$



$$f(x) + f'(x) \rightarrow f''(x) < 0 \rightarrow \text{maximizing}$$



$$\Rightarrow x^2 - x + 2 = f(x)$$

$$f'(x) = 2x - 1 \rightarrow$$

$$f''(x) = 2$$

$$2x - 1 = 0$$

$$2x = 1$$

$$x = 1/2$$

↓

mining

$$\Rightarrow p(x) = 41 - 72x - 10x^2$$

Max profit?

$$p'(x) = 0 - 72 - 36x$$

$$p''(x) = -36 \rightarrow \text{maximal}$$

$$-72 - 36x = 0$$

$$x = -2$$

$$p(x) = 41 - 72(-2) - 18(-2)^2 \\ = 113$$

$\Rightarrow$

$$y = x^2 + 7 \rightarrow \text{Pak mittle}$$

$$(3, 7)$$

$$(x, x^2 + 7)$$

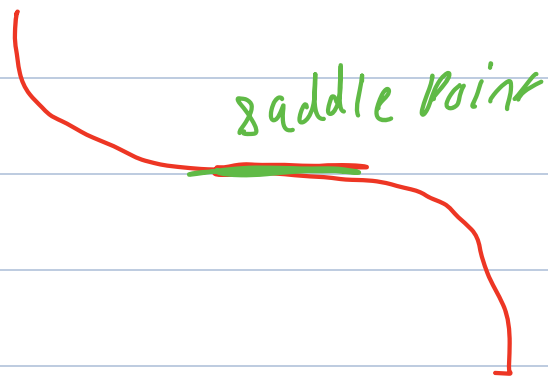
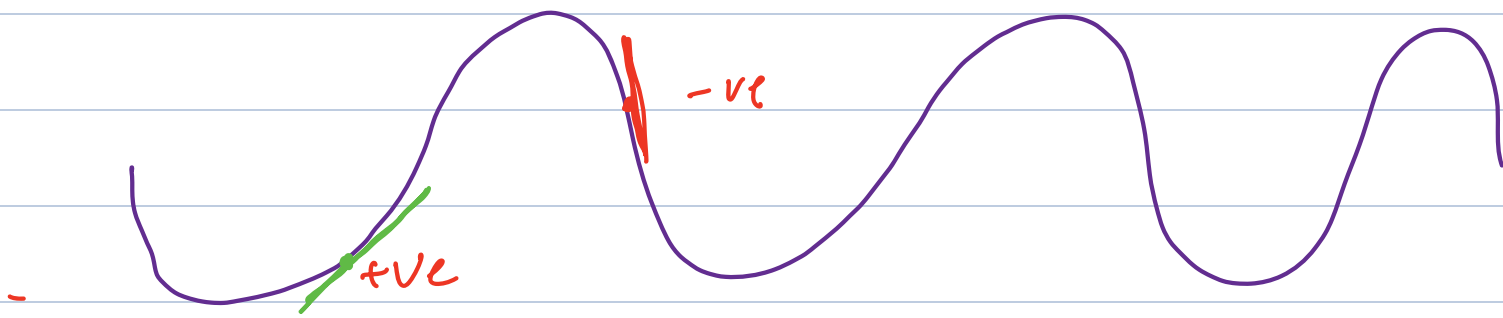
$$(3, 7)$$

$$\sqrt{(x-3)^2 + (x^2+7-7)^2}$$

$$\sqrt{(x-3)^2 + x^4} \rightarrow f(x)$$

$$\Rightarrow (x-3)^2 + x^4 = 0$$

$$\downarrow \\ \sqrt{5}$$



$\Rightarrow$

$$\max_{\vec{w}, w_0} \sum_{i=1}^n y_i \cdot \frac{w^T x_i + w_0}{||w||}$$

Loss Func

$$\max \rightarrow 20, 50 \Rightarrow 50$$

$$\min \rightarrow -20, -50 \Rightarrow -50$$

$$\min_{\vec{w}, w_0} - \sum_{i=1}^n y_i \cdot \frac{w^T x_i + w_0}{\|w\|}$$

$$f(x) = 2x^2 + 4x + 3$$

$$y = f(x_1, x_2, x_3, \dots, x_n)$$

$$\frac{df(x_1, x_2, x_3, \dots, x_n)}{dx} \Rightarrow ?$$

Age | wght | hght | salary

$$wght = 2 * Age + 3 * hght$$

$$x_1, x_2$$

$$x_2 = f(x_1)$$

Partial Derivative

$$x_1, x_2, x_3$$

$$f(x, y) = 2x^2y + 3y^3x^2 + 3y$$

$$\frac{\partial f(x, y)}{\partial x} = 4xy + 6y^3x$$

$$\frac{\partial f(x, y)}{\partial y} = 2x^2 + 9y^2x^2 + 3$$

Goal :

$$\min_{w_0, w_1, w_2, \dots} f(w_0, w_1, w_2, \dots, w_n)$$

$$\frac{\partial f}{\partial w_0}, \frac{\partial f}{\partial w_1}, \frac{\partial f}{\partial w_2}, \dots$$

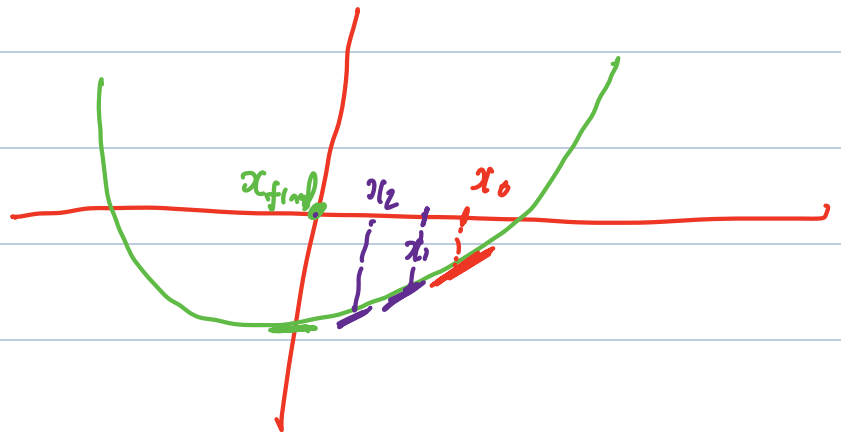
$$\nabla f = \begin{bmatrix} \frac{\partial f}{\partial w_0} \\ \frac{\partial f}{\partial w_1} \\ \vdots \\ \frac{\partial f}{\partial w_n} \end{bmatrix} \rightarrow \text{Gradient}$$

Gradient Descent Intuition

$$y = x^2 - 30$$

$$f'(x) = 2x$$

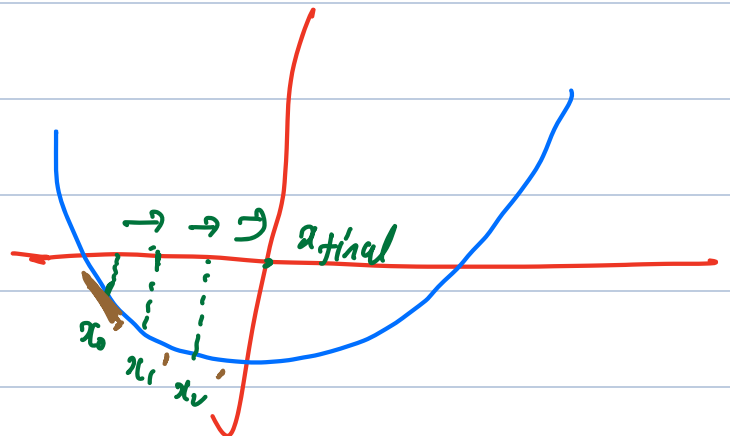
$$f'(x_0) = +ve$$

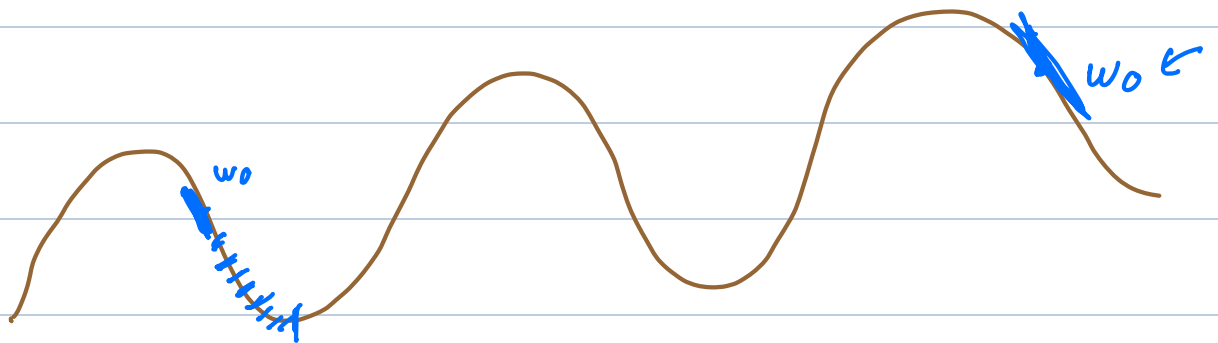


$$x_1 = x_0 - \eta f'(x_0)$$

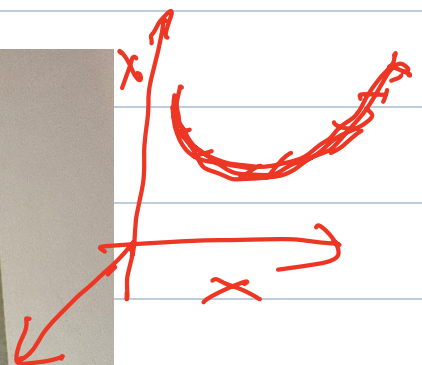
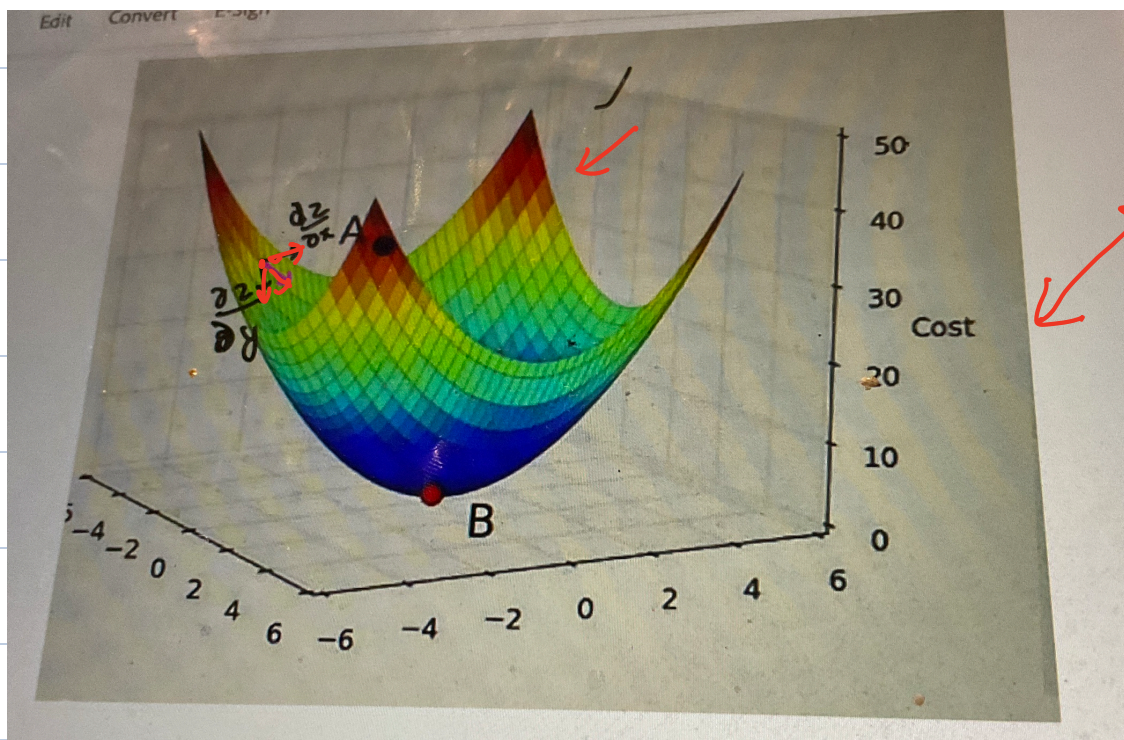
$$x_1' = x_0' - \eta f'(x_0')$$

↓
learning rate





$$w_{\#} = w_0 - \underbrace{f'(w_0)}_{\text{ve}}$$

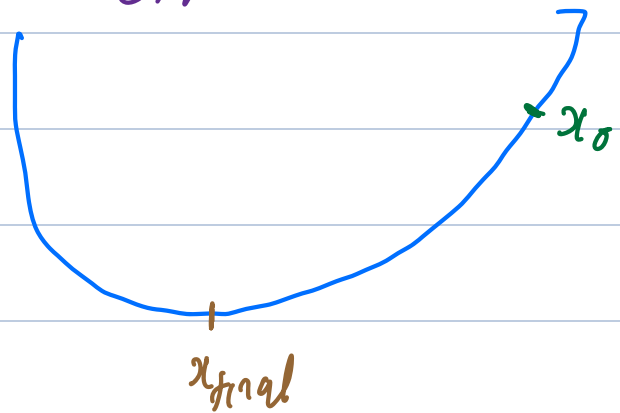
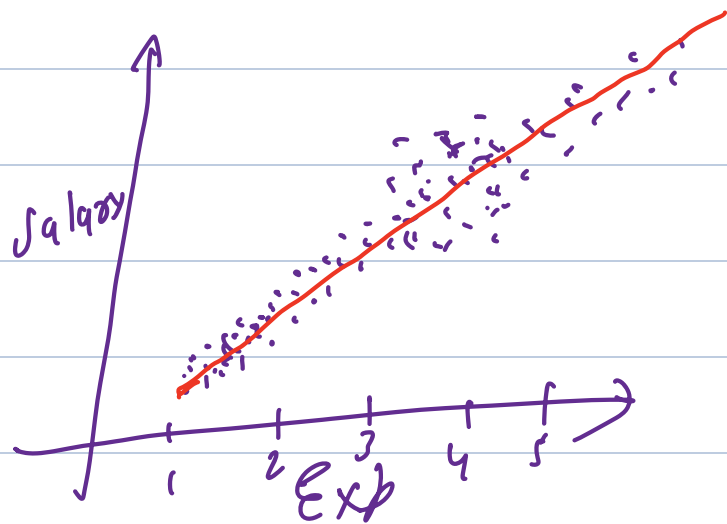
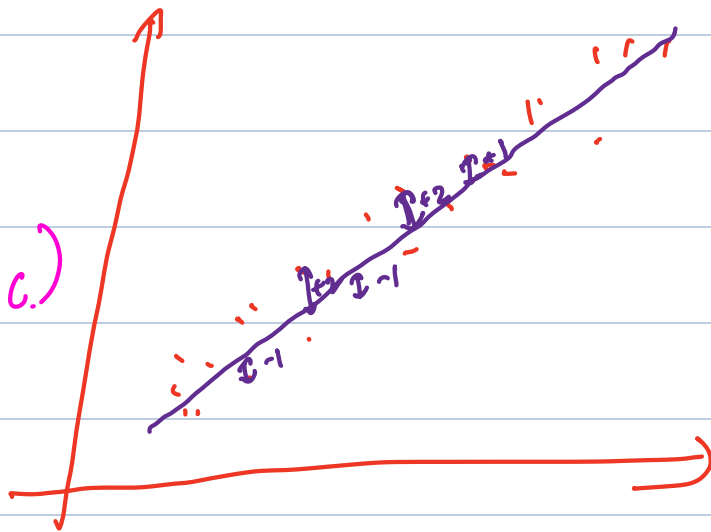


$$\hat{y}_i = mx_i + c$$

$$\sum_{i=1}^n (y_i - \hat{y}_i)^2 = J(m, c)$$

$$(-2)^2 + (-3)^2 + (2)^2 + (-3)^2$$

$$\approx 0$$

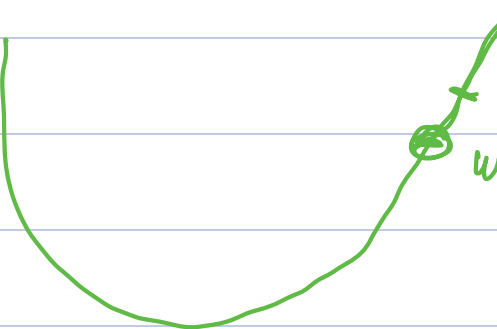


$$\sum_{i=1}^n (y_i - \hat{y}_i)^2 = J(m, c)$$

$$\sum_{i=1}^N \frac{1}{N} (y_i - (mx_i + c))^2 = J(m, c)$$

$$\frac{\partial J}{\partial m} = \frac{2}{N} \sum (y_i - (mx_i + c)) (-x_i)$$

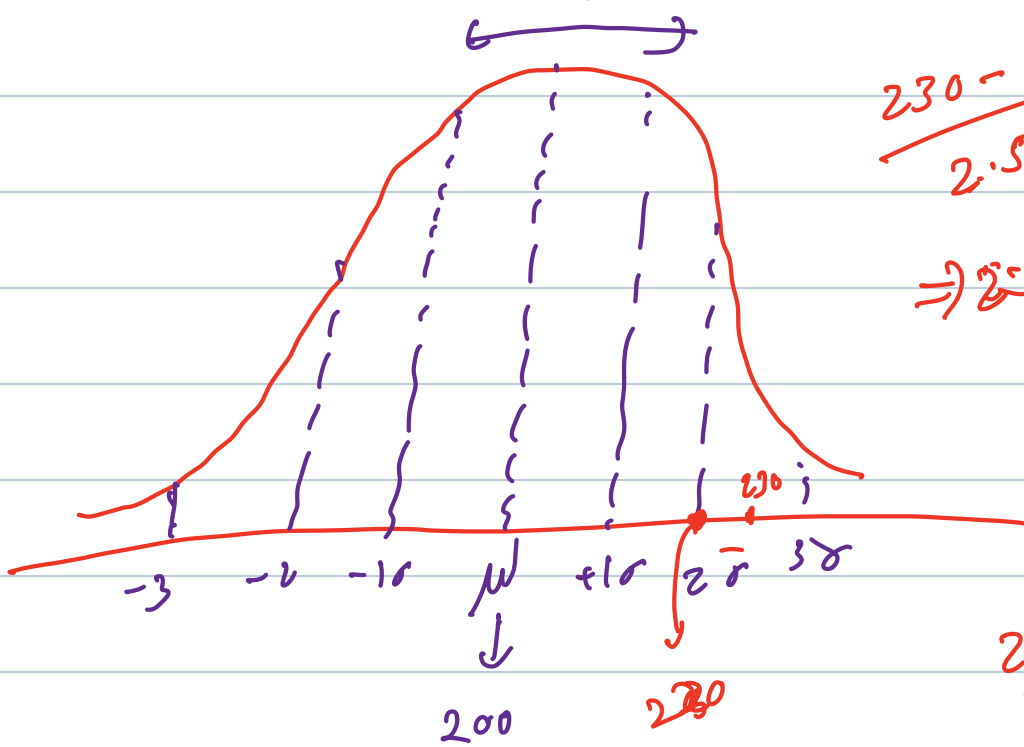
$$\frac{\partial J}{\partial c} = \frac{2}{N} \sum (y_i - (mx_i + c)) (-1)$$

$$w_1[\text{next}] \Rightarrow w_1[\text{current}] - \eta \frac{\partial f}{\partial w_1}$$


$$w_1 = w_1 - \eta \frac{\partial f}{\partial w_1}$$

$$w_n[\text{next}] = w_n[\text{current}] - \eta \frac{\partial f}{\partial w_n}$$

$$\begin{array}{c} 99\% \\ \hline 95\% \\ \hline 65\% \end{array}$$



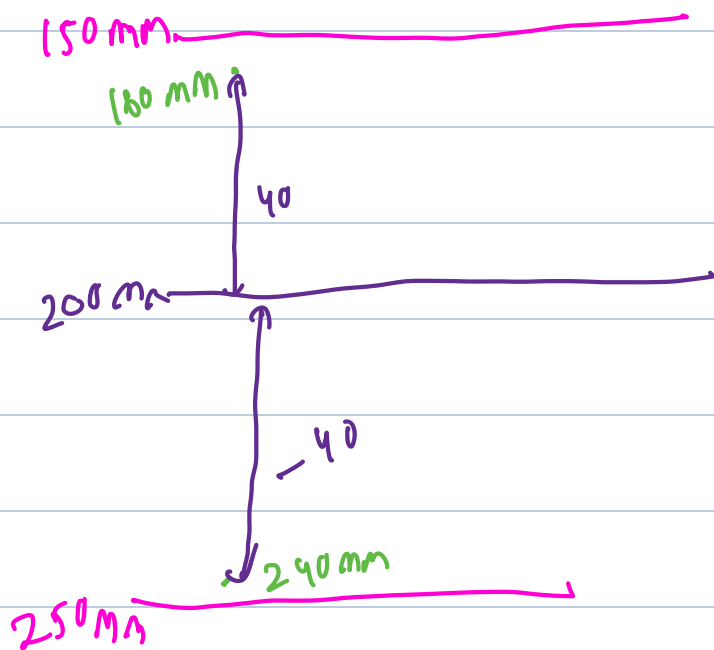
$$\frac{230 - 200}{2.5\sigma}$$

$$\Rightarrow 2.5$$

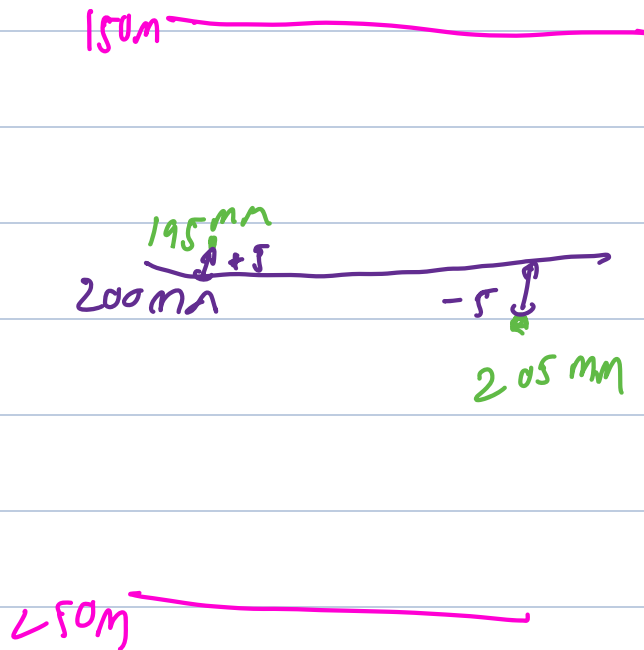
$$\frac{220 - 200}{2 \times 10}$$

$$\sigma = 10$$

M1



M2



$$\text{variance} = \frac{(40)^2 + (-40)^2}{2}$$

$$\frac{3200}{2} \Rightarrow \boxed{1600}$$

$$\frac{(5)^2 + (-5)^2}{2}$$

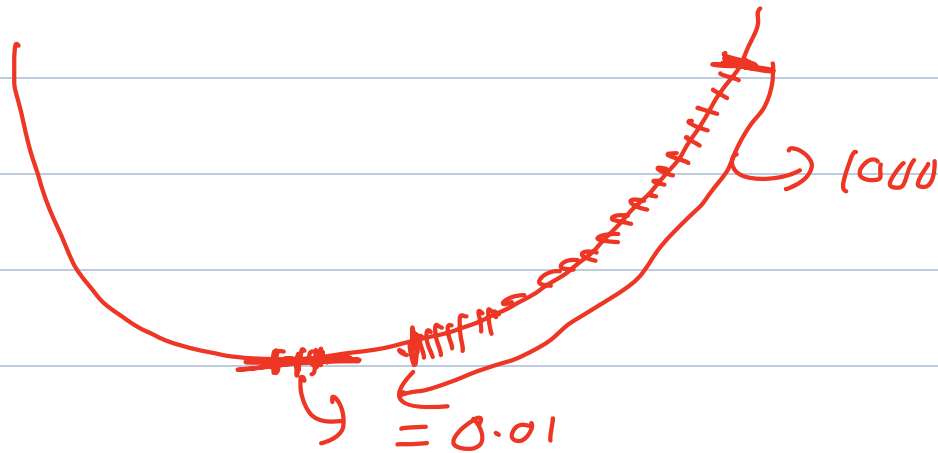
$$\boxed{25}$$

$$Jd = \sqrt{Uq\delta}$$

```
# Implement gradient descent function
# Takes in X, y, current m and c (both initialised to 0), num_iterations, learning rate
# returns gradient at current m and c for each pair of m and c
```

```
def gradient(X, y, m_current=0, c_current=0, iters=1000, learning_rate=0.01):
    N = float(len(y))
    gd_df = pd.DataFrame( columns = ['m_current', 'c_current', 'cost'])
    for i in range(iters):
        y_current = (m_current * X) + c_current
        cost = sum([data**2 for data in (y-y_current)]) / N
        m_gradient = -(2/N) * sum(X * (y - y_current))
        c_gradient = -(2/N) * sum(y - y_current)
        m_current = m_current - (learning_rate * m_gradient)
        c_current = c_current - (learning_rate * c_gradient)
        gd_df.loc[i] = [m_current, c_current, cost]
    return(gd_df)
```

$$Error = \sum_{i=1}^N \frac{1}{N} (y - \hat{y})^2$$



```
[ ] # print gradients at multiple (m, c) pairs
# notice that gradient decreased gradually towards 0
# we have used 1000 iterations, can use more if needed
gradients = gradient(X,y)
gradients
```



	m_current	c_current	cost
0	0.010700	3.731979e-18	0.998165
1	0.021187	1.981697e-17	0.986830
2	0.031464	2.998621e-17	0.975943
3	0.041536	2.656387e-17	0.965486
4	0.051407	2.433120e-17	0.955442
5	0.061081	2.840541e-17	0.945796
6	0.070562	3.605678e-17	0.936530

Area	Price
2	20K
3	30K
4	40K

$X = [2, 3, 4]$

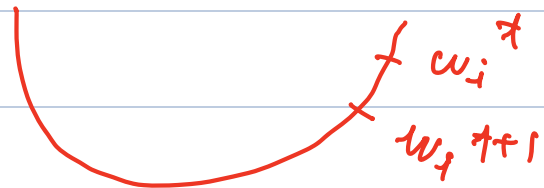
~~Price~~
 $y = [20K, 30K, 40K]$

Recap:

$$\min_{w_0, \vec{w}} - \sum y_i \cdot \frac{w^T x_i + w_0}{\|w\|}$$

GD:

$$w_i^{*+1} = w_i^* - \eta \frac{\partial f}{\partial w_i^*}$$



$$\Rightarrow - \sum y_i \cdot \frac{w_1 x_1 + w_2 x_2 + \dots + w_0}{\sqrt{w_1^2 + w_2^2 + w_3^2 + \dots + w_n^2}}$$

$\rightarrow f(x)$
 $\rightarrow g(x)$

$$- \sum y_i \cdot \frac{f(x)}{g(x)}$$

$\therefore$ computationally expensive

$$\underbrace{3x + 4y + 4}_{\substack{\downarrow w_1 \\ \|w\|}} = 0$$

$$\|w\| = \sqrt{w_1^2 + w_2^2}$$

$$= \sqrt{3^2 + 4^2} = \sqrt{25} = 5$$

$$\underbrace{\left(\frac{3}{5}\right)}_{w_1'} x + \underbrace{\left(\frac{4}{5}\right)}_{w_2'} y + \underbrace{\left(\frac{4}{5}\right)}_{w_0'} = 0$$

$$w_1' = \frac{w_1}{\|w\|}$$

$$w_2' = \frac{w_2}{\|w\|}$$

$$w_0' = \frac{w_0}{\|w\|}$$

$$\begin{aligned} \min_{\vec{w}, w_0} \quad & - \sum y_i (w^T x + w_0) \\ \text{s.t.} \quad & \underbrace{\|w\| = 1}_{\text{condition}} \end{aligned}$$

$$\frac{\partial f}{\partial w_1} \Rightarrow \frac{d}{dw_1} - \sum y_i (w_1 x_1 + w_2 x_2 + \dots + w_n) \\ = - \sum y_i x_i$$

$$w_j^{(t+1)} = w_j^{(t)} - \eta \left(- \sum_{i=1}^n y_i x_i \right)$$

$$\frac{\partial f}{\partial w_0} = - \sum y_i$$

$$w_0^{(t+1)} = w_0^{(t)} - \eta (- \sum y_i)$$

General form

min/max $\rightarrow$ optimisation problem $\rightarrow f(\theta)$

s.t.

constraint fun. $\rightarrow g_1(\theta), g_2(\theta)$
 $\downarrow$

$$\|w\| = 1$$

$\Downarrow$

$$\|w\| - 1 = 0$$

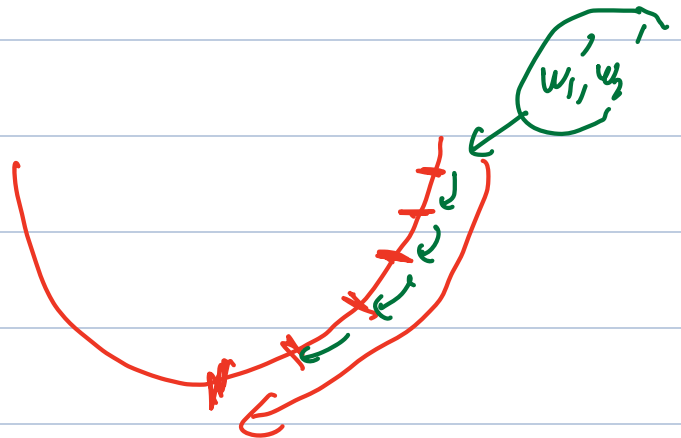
$$\min_{w, w_0} - \sum y_i \cdot (w^T x_i + w_0)$$

$$\text{s.t. } \|w\| - 1 = 0$$

GD ?

η

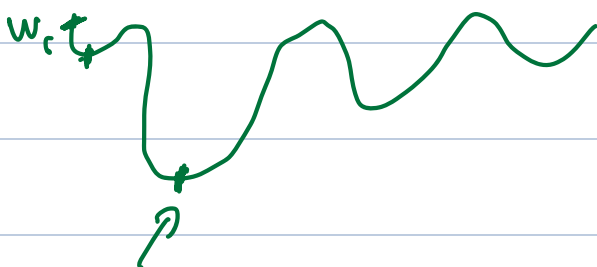
$$w_1' = w_1 - \frac{\partial f}{\partial w_1}$$



Netflix

$\Leftarrow$

Bat



Lagrange Multiplier

Constraint prob $\rightarrow$ Unconstraint prob

$$\min_{w_0, \vec{w}} - \sum y_i \cdot \frac{w_1 x_1 + w_2 x_2 + \dots + w_0}{\|\vec{w}\|} \rightarrow \text{unconstraint prob}$$

$\Downarrow$

$$\min_{w_0, \vec{w}} - \sum y_i (w^T x + w_0)$$

$\rightarrow$ constraint prob

s.t. $\|\vec{w}\| = 1$

$\Downarrow$

unconstraint problem $\rightarrow$ easier to solve

$$\min f(\theta), \text{ s.t. } g_1(\theta) = 0 \rightarrow \|\vec{w}\| - 1 = 0$$
$$g_2(\theta) = 0$$
$$g_3(\theta) = 0$$

η

$\downarrow$ learning rate

$$\min_{\theta} f(\theta) + \lambda_1 g_1(\theta) + \lambda_2 g_2(\theta) + \lambda_3 g_3(\theta)$$

Lagrange multiplier

Ex)

$$\min_{x,y} x^2 + y^2$$

$$\text{s.t. } x + 2y - 1 = 0$$

$\Downarrow$

$$\min_{x,y,\lambda} x^2 + y^2 + \lambda (x + 2y - 1) = L$$

$$\frac{\partial L}{\partial x} = 2x + \lambda \Rightarrow \lambda = -2x$$

$$\frac{\partial L}{\partial y} = 2y + 2\lambda \Rightarrow \lambda = -y$$

$$\frac{\partial L}{\partial \lambda} = x + 2y - 1$$

$$x + 2y - 1 = 0$$

$$x + 2(2x) - 1 = 0$$

$$x = 1/5 \quad y = 2/5$$

$$\lambda = -\frac{2}{5}$$

$$0 \Rightarrow \min_{\vec{w}, w_0} -\sum y_i (w^T x_i + w_0)$$

s.t. $\|w\| - 1 = 0$

↓

$$\mathcal{L} = \min_{\vec{w}, w_0} -\sum y_i (w^T x_i + w_0) + \lambda (\|w\| - 1)$$

Break: 9:21 PM